

RESEARCH SUMMARY - Qwen3-VL-30B-A3B Expert Routing Analysis

Model : Qwen/Qwen3-VL-30B-A3B-Instruct
Quantisation : 4-bit NF4 (bitsandbytes)
MoE layers : 48
Experts / layer : 128 (top-4 activated)
Sequence length : 2768 tokens
Image tokens : 2754 (99.5%)
Text tokens : 14 (0.5%)

--- Divergence Metrics ---

KL(image || text) : 0.0213
KL(text || image) : 0.0217
JS divergence : 0.0215

--- Expert Overlap (top-10) ---

Shared : 0/10
Image-specialised : 10
Text-specialised : 10

--- Entropy (bits, averaged across layers) ---

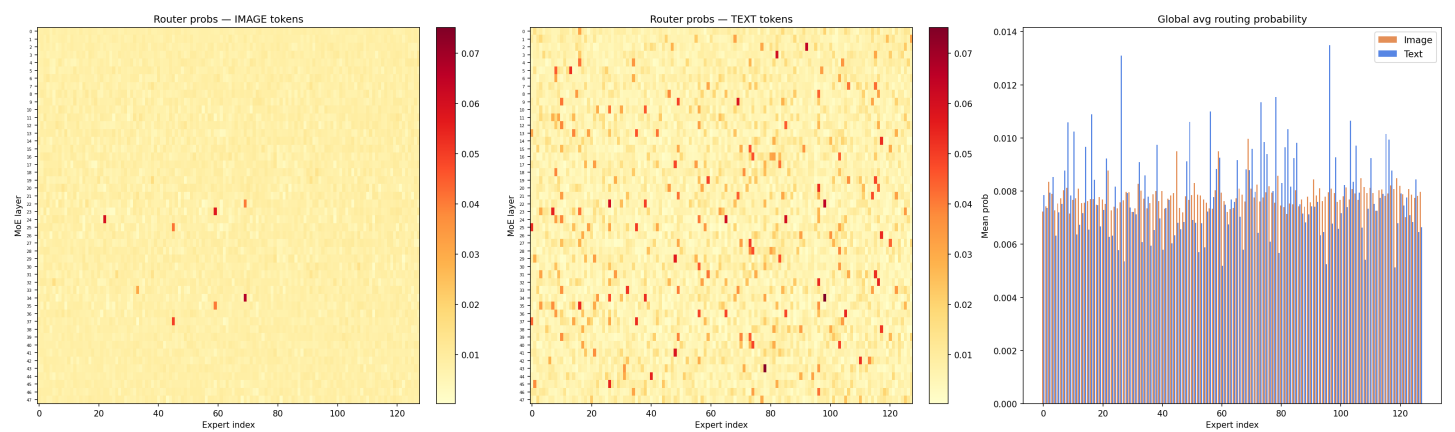
Image tokens : 6.954
Text tokens : 6.589

Finding: Text tokens exhibit LOWER entropy (more concentrated routing), suggesting text processing relies on a narrower set of preferred experts.

100% of the top-10 image experts are NOT in the top-10 text experts, indicating modality-specific expert specialisation.

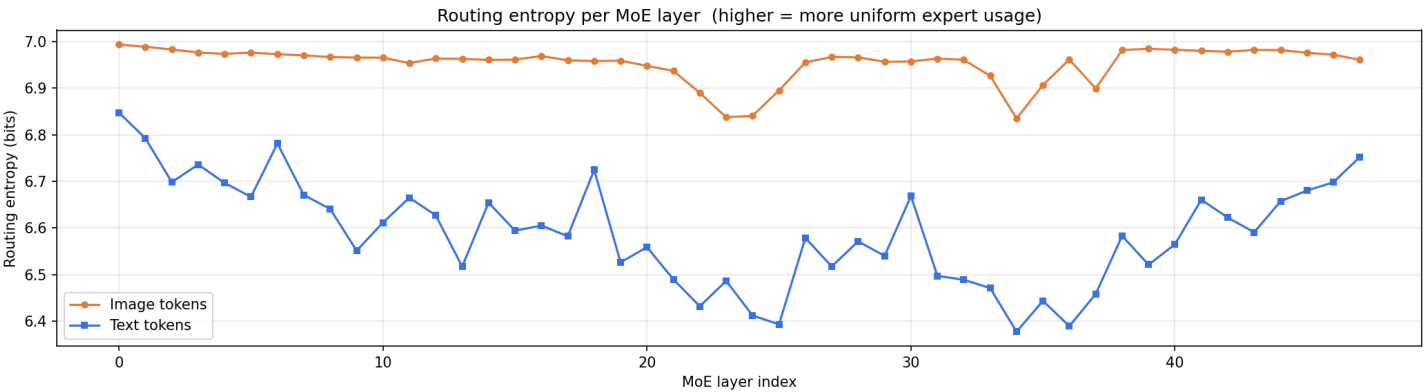
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Global Routing Heatmaps



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Routing Entropy per Layer



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Token-level Routing Probabilities

